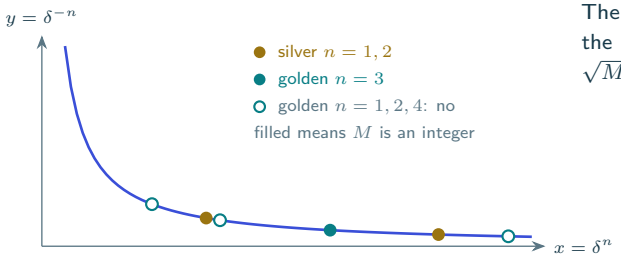
NUMBER THEORY ON ONE PAGE

Metallic means as differences of squares

When is $\delta_k^{-n} = \sqrt{M+1} - \sqrt{M}$ for an integer M? Every n if k is even, only multiples of 3 if k is odd.

1 / THE RULE AND THE PICTURE

Powers of $\delta_k = \frac{k+\sqrt{k^2+4}}{2}$ lie on the hyperbola $xy = 1$



The metallic mean δ_k is the positive root of $x^2 - kx - 1 = 0$, so $\delta_k^{-1} < 1$ and the powers δ_k^{-n} shrink towards 0. The question is when $y = \delta_k^{-n}$ can be written $\sqrt{M+1} - \sqrt{M}$ with $M \in \mathbb{Z}_{\geq 0}$.

$$M_n(k) = \frac{\delta_k^{2n} + \delta_k^{-2n} - 2}{4} = \frac{S_n - 2}{4}$$

2 / THE MECHANISM / RATIONALISE, THEN SQUARE

The formula for M comes from y and $1/y$, and S_n is a Lucas sequence

If $y = \sqrt{M+1} - \sqrt{M}$ then $1/y = \sqrt{M+1} + \sqrt{M}$, so $1/y - y = 2\sqrt{M}$ and $M = \left(\frac{1/y - y}{2}\right)^2 = \frac{y^{-2} + y^2 - 2}{4}$. With $y = \delta_k^{-n}$ this gives $M_n(k)$, and since $\delta_k^2 + \delta_k^{-2} = k^2 + 2$,

$$S_{n+1} = (k^2+2)S_n - S_{n-1}, \qquad S_0 = 2,\ S_1 = k^2 + 2, \qquad M_n \in \mathbb{Z} \iff S_n \equiv 2 \pmod 4.$$

S_n	$n=0$	1	2	3	4	5	6
$k=1$	2	3	7	18	47	123	322
$k=2$	2	6	34	198	1154	6726	39 202
$k=3$	2	11	119	1298	14 159	154 451	1 684 802

3 / THE PARITY THEOREM / WORKING MODULO 4

Even k always works, and odd k works exactly when $3 \mid n$

Put $A = k^2 + 2$. For even k, $A \equiv 2$ and $S_{n+1} \equiv 2S_n - S_{n-1} \pmod 4$ from $S_0 \equiv S_1 \equiv 2$ keeps $S_n \equiv 2$ for ever. For odd k, $A \equiv 3$ and $S_{n+1} \equiv -S_n - S_{n-1}$, which cycles with period three:

n	0	1	2	3	4	5	6	7	8
$S_n \bmod 4$, k even	2	2	2	2	2	2	2	2	2
$S_n \bmod 4$, k odd	2	3	3	**2**	3	3	**2**	3	3

k even: $M_n(k) \in \mathbb{Z}\ \forall n$; $\quad$ k odd: $M_n(k) \in \mathbb{Z} \iff 3 \mid n$

4 / WHY PERIOD THREE / THE CUBE TRICK AND SQUARES

Cubing turns an odd metallic mean into an even one

Cubing gives $\delta_k^3 = \delta_{k^3+3k}$, and $k^3 + 3k = k(k^2+3)$ is even exactly when k is odd: $\varphi^3 = 2 + \sqrt{5} = \delta_4$ and $\delta_3^3 = 18 + \sqrt{325} = \delta_{36}$. An even-index mean has $\delta_j^{-1} = \sqrt{(j/2)^2 + 1} - \frac{j}{2}$, so $\varphi^{-3} = \sqrt{5} - \sqrt{4}$ and $\delta_3^{-3} = \sqrt{325} - \sqrt{324}$. Equivalently $M_n = \left(\frac{V_n}{2}\right)^2$ for odd n and $\frac{DU_n^2}{4}$ for even n, with $D = k^2 + 4$ and U_n, V_n the Lucas sequences, and these are integers when k is even or when U_n, V_n are even, which for odd k happens exactly when $3 \mid n$. For $k = 2$ this gives $M = 1,\ 8,\ 49,\ 288$: alternately a square and twice a square.

5 / EXAMPLES / THE REPRESENTATIONS

Silver always, golden and bronze at multiples of three

mean	power	difference of square roots	M
golden, $k=1$	φ^{-3}, φ^{-6}	$\sqrt{5} - \sqrt{4}$, $\sqrt{81} - \sqrt{80}$	4, 80
silver, $k=2$	$(\sqrt{2}-1)^{1,2,3,4}$	$\sqrt{2} - \sqrt{1}$, $\sqrt{9} - \sqrt{8}$, $\sqrt{50} - \sqrt{49}$, $\sqrt{289} - \sqrt{288}$	1, 8, 49, 288
bronze, $k=3$	δ_3^{-3}, δ_3^{-6}	$\sqrt{325} - \sqrt{324}$, $\sqrt{421\,201} - \sqrt{421\,200}$	324, 421 200
$k=4$	δ_4^{-1}	$\sqrt{5} - \sqrt{4}$ (the golden cube)	4

Numerically $\varphi^{-3} = 0.23607 = \sqrt{5} - 2$ and $(\sqrt{2}-1)^4 = 0.029437 = \sqrt{289} - \sqrt{288}$.

ASIDE / THE GALOIS PICTURE

δ_k is a unit of norm -1 with conjugate $-\delta_k^{-1}$, and modulo 4 the odd-k recurrence has characteristic polynomial $x^2 + x + 1$, whose roots are cube roots of unity.

ASIDE / CHECK IN PYTHON

Iterating `a,b = b,(k*k+2)*b-a` from `(2,k*k+2)` and testing `(S-2)%4==0` reproduces the table.

Convention. $S_n = \delta_k^{2n} + \delta_k^{-2n}$; for $k = 1$ these are the Lucas numbers L_{2n}. After the Metallic Means: Consecutive Square Roots page.